FLICKRING

TOUCH-FREE PHONE CONTROL

Control your phone with a small ring ,even with wet,gloved or unsteady hands



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FlickRing: Touch-Free Phone Control

Problem

- Touchscreens fail with gloves, water and tremor
- Voice fails in noisy kitchens and garages
- Gesture wearables are costly and locked to one brand

Innovation

- Motion-sensing ring
- Learns each user
- Wake gesture stops accidental actions
- Open app: map gestures to any task

Solution

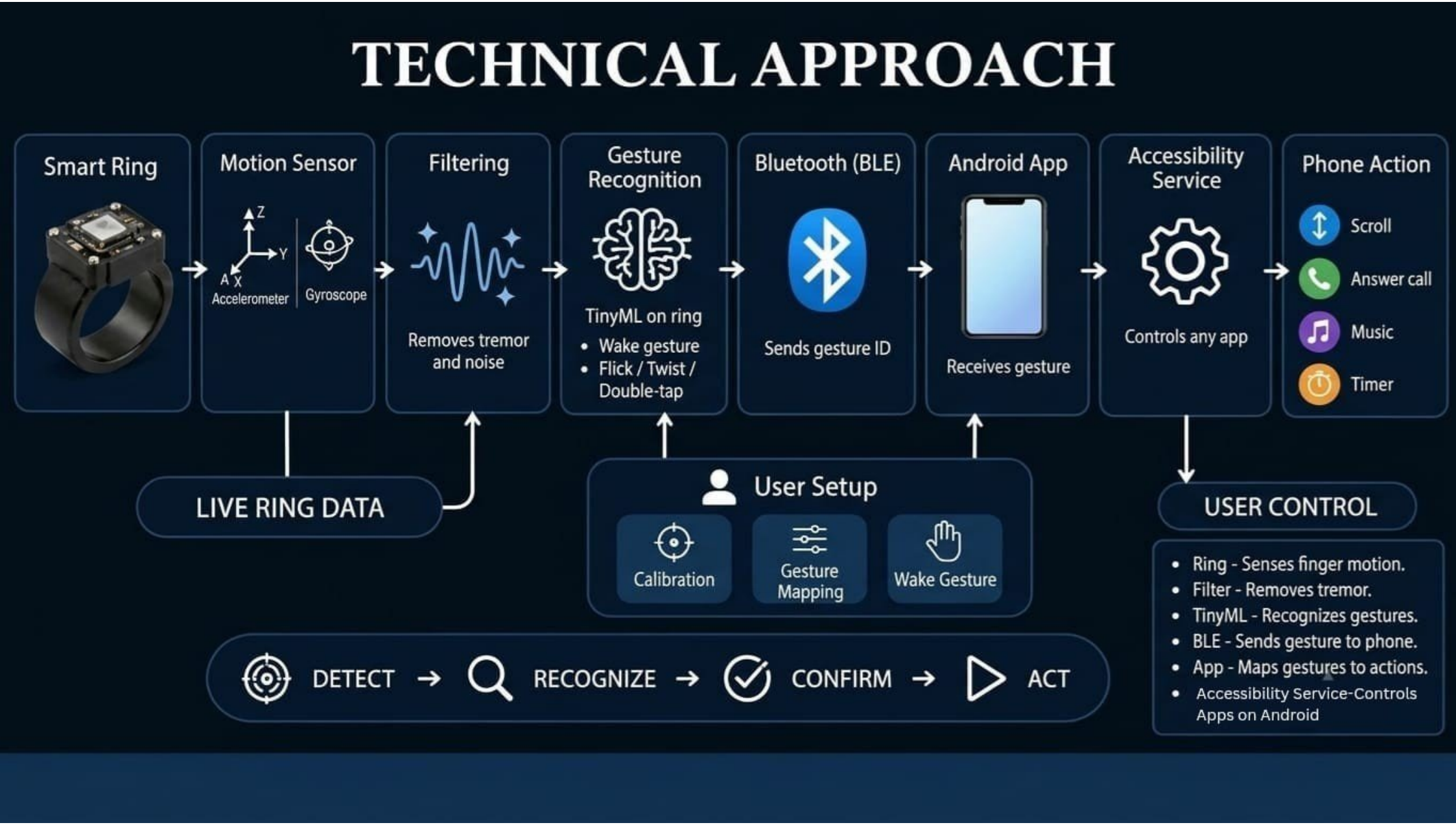
1. **SENSE:** 3D-printed ring with XIAO nRF52840 Sense board and filter
2. **DECIDE:** on-device model recognizes flick, double-tap and twist
3. **ACT:** Bluetooth to Android app: next step, scroll, answer call

Need: affordable phone control without touch or voice

Core Technical Flow

Finger motion → Ring sensor → Gesture recognition → Bluetooth → Phone app → Action

TECHNICAL APPROACH



Feasibility and Viability

FEASIBILITY

1. **Technical**
   IMU + BLE chip (XIAO Sense)
2. **Economic**
   Low-cost, available parts
3. **Design**
   Wearable ring, 3D-printed case
4. **Implementation**
   Embedded code + recognition
5. **Operational**
   Remappable, simple gestures
6. **Testing**
   Accuracy, speed, battery life

VIABILITY

Market Potential
Growing interest in wearables

Cost-Effective
Affordable build, compact parts

Convenient
Intuitive, easy-to-learn gestures

Product Potential
Can grow into a product with demand

Impact and Benefit

KEY BENEFITS

1. **Touch-Free Control**
   Scroll, answer calls, no screen touch
2. **Works in Noise**
   Where voice commands fail
3. **Personalized Gestures**
   Users map gestures to fit each hand
4. **Accessible Design**
   Option for limited mobility or tremor
5. **Affordable and Open**
   Alternative to costly, closed wearables

REAL-WORLD IMPACT

1. **Fewer Interruptions**
   Chefs, mechanics, lab technicians
2. **Cleaner Screens**
   Less contact with dirty, wet fingers
3. **Greater Independence**
   Elderly and mobility-challenged users
4. **Affordable Assistive Tech**
   Lower cost, pending cost validation
5. **Wider Use**
   Smart-home, touch-free navigation

Research and References

WHY CURRENT OPTIONS FAIL

1. Touch: wet fingers, gloves disrupt screens [7]
2. Voice: noise degrades recognition [8]
3. Tremor: harder to hit touch targets [5][6]

WHY A RING CAN WORK

1. IMU ring: 95.5% on 15 gestures [1]
2. VibRing: 94.2% accuracy reported [4]
3. Ring gestures can control a phone [2] [3]

Gap (to our knowledge): low-cost, open ring for wet, gloved, shaky hands

References

[1] Liu et al., 2020 - IMU ring, IMWUT
[2] Liang et al., 2021 - DualRing, IMWUT
[3] Wolf et al., 2016 - Microgestures, MobileHCI
[4] Li et al., 2025 - VibRing, PACM HCI
[5] Wacharamanotham et al., 2011 - Swabbing, CHI
[6] Dunlop and MacKenzie 2026 - Tremor, Sensors
[7] Tung et al., 2018 - RainCheck, ICMI
[8] Barfuss et al., 2016 - Speech in noise, arXiv